

The nearly-free-electron model에 대한 고찰

Contents

1. Introduction : Lattice and Basis
 - 1.1 Lattice and Basis
 - 1.2 The Brillouin Zones and Fermi Surface
2. The nearly-free-electron model
 - 2.1 Perturbation Theory
 - 2.2 The Schrödinger equation in a periodic potential with Hamiltonian
 - 2.3 Degenerated Perturbation Theory
3. Vanishing potential
 - 3.1 Single electron energy state
 - 3.2 Several degenerate energy levels
 - 3.3 Two degenerate free-electron levels

1. Introduction : The nearly-free electron model

In Chapter 2, we learned about k -space and Bloch's theorem. Through these concepts, we solved the Schrödinger equation.

However, what is needed in this model, we need to learn what is called perturbation theory, which we learn at a level above quantum physics 2, so we will proceed in this chapter with a brief introduction and summary.

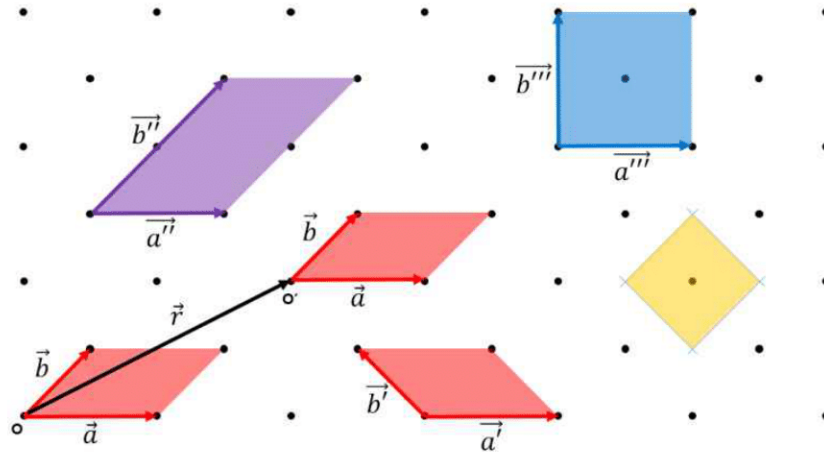
A brief summary for The nearly-free electron model is as follows :

Potential is treated as a very small perturbation, and Perturbation theory is used, and using Perturbation theory results in a bandgap at the Brillouin zone boundary. We have to watch up to the second order.

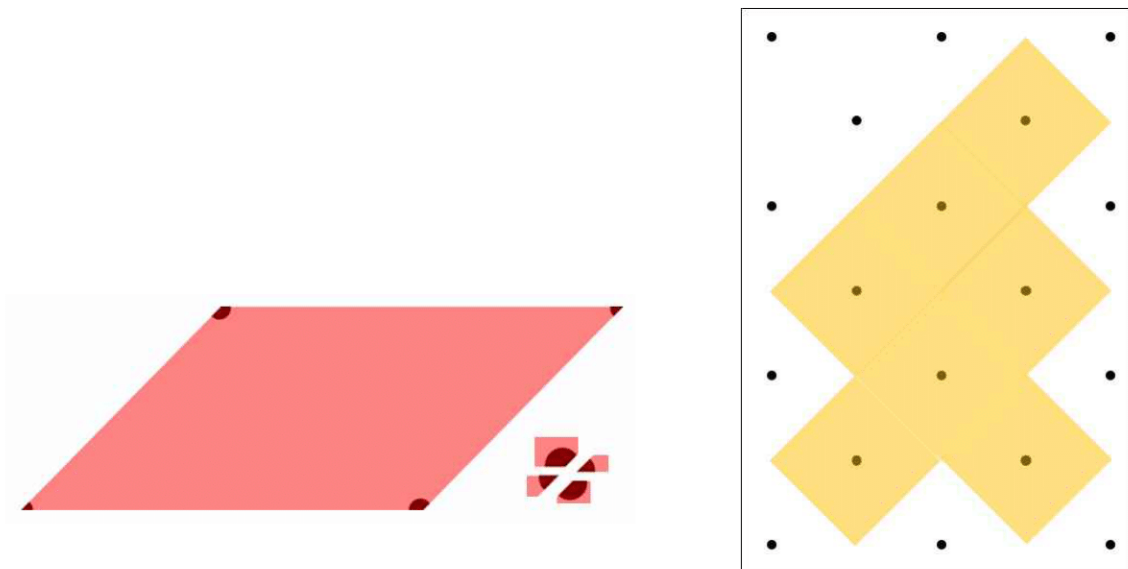
There is a degenerate in the Brillouin zone boundary, there is an energy region where electrons cannot exist because of the bandgap, and the bandgap opens at the Brillouin zone boundary by the periodic potential in the solid.

1-1 Lattice and Basis

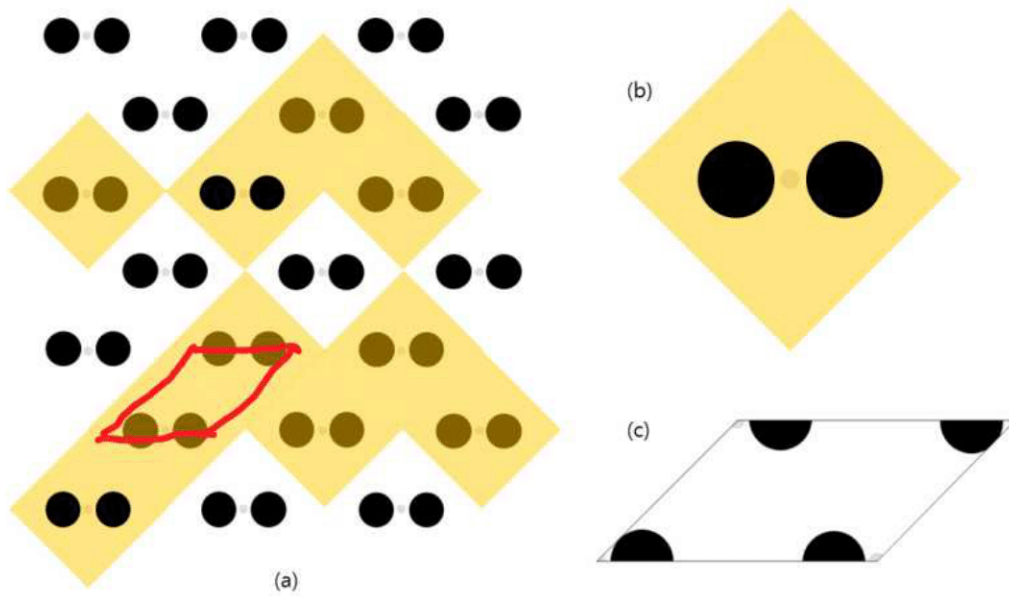
The Lattice is 'an infinite repetition', while the Basis is 'a group of atoms to be placed equally in Lattice' This is visualized in two dimensions as follows.



The dots of each of picture is Lattice points, and the arrow from o to o' are translation vector, the connection of the nearest Lattice point, such as vector a and vector b , is called the Primitive vector.



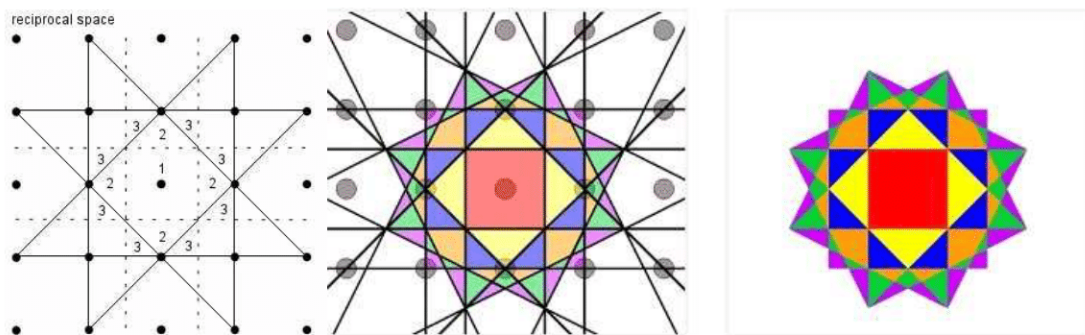
The left picture is a visualization of a Primitive unit cell consisting of a Primitive vectors, the Primitive unit cell contains only one Lattice point, the right picture means that by setting up a small unit cell and connecting the cell indefinitely, you can fill the Lattice Space, the cell shape in the right picture is called a Wigner-Seitz cell.



The basis can be easily understood visually as a picture of a Wigner–Seitz cell with a Basis with one atom placed on each side of the Lattice point as a Primitive unit cell.

1.2 The Brillouin Zones and Fermi Surface

The Brillouin Zone



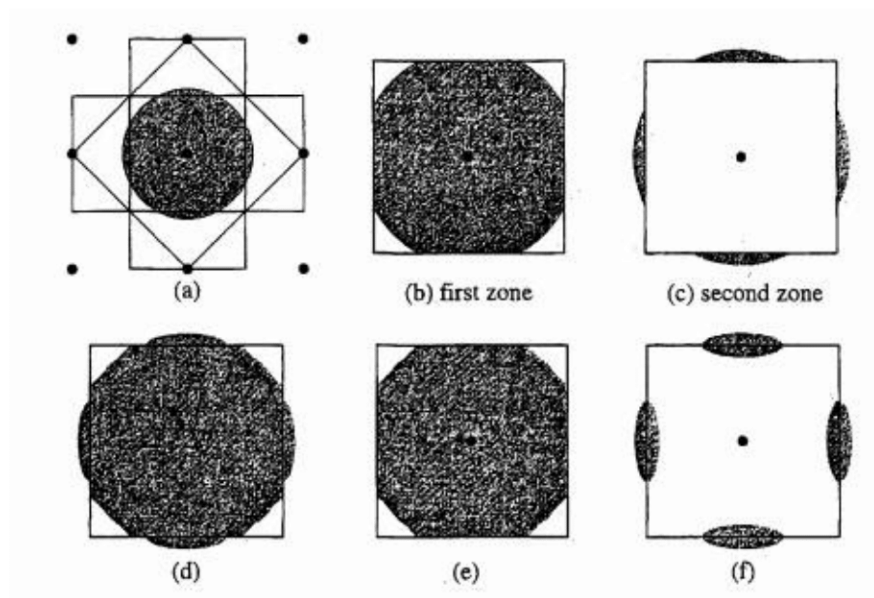
The set of zones that occur when atoms are bisected in K-space is called the Brillouin zone. The first unit-cell surrounding the reference atom is called the 1st Brillouin zone, and surrounding the 1st Brillouin zone is called the 2nd Brillouin zone. This is how the 3rd, 4th... Brillouin zone repeats itself.

And what's important is that each color has the same width.

That means that the area of the 1st brillouin zone and the area of the 2nd, 3rd...brillouin zone are all the same, which means that the information that the 1st brillouin zone has is the same as the rest of the brillouin zone.

This is called a reduction region method in which a portion deviating from the first brillouin band is returned to the brillouin band, that is, reduced (reduced zone representation).

The evolution of the Fermi surface of a divalent two-dimensional metal with a square lattice as a band gap is opened at the Brillouin zone boundary :



(a) is free-electron Fermi surface (shaded circle), reciprocal lattice points (solid dots) and first (square) second (four isocetes triangles) and third (eight isocetes triangles) Brillouin zones;

(b) is distortion of the Fermi surface enclosed by the first Brillouin zone boundaries;

(c) is the sections of Fermi surface in the second Brillouin zone;

(d) is distortion of the Fermi surface section shown in (a) due to formation of band gaps at the Brillouin zone boundaries;

(e) is the section of the distorted Fermi surface enclosed by the first Brillouin zone;

(f) is result of the distortion of the Fermi surface section in (c) plus 'folding back' of these sections due to the periodicity of k -space.

The Fermi Surface

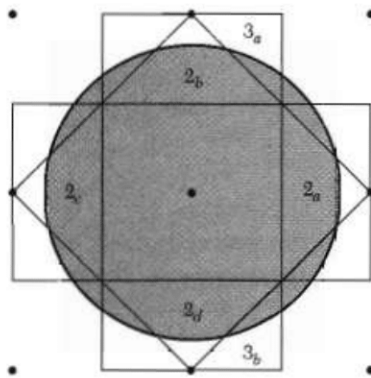


Figure 6 Brillouin zones of a square lattice in two dimensions. The circle shown is a surface of constant energy for free electrons; it will be the Fermi surface for some particular value of the electron concentration. The total area of the filled region in k space depends only on the electron concentration and is independent of the interaction of the electrons with the lattice. The shape of the Fermi surface depends on the lattice interaction, and the shape will not be an exact circle in an actual lattice. The labels within the sections of the second and third zones refer to Fig. 7.

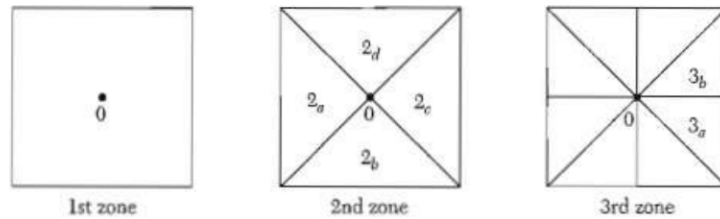


Figure 7 Mapping of the first, second, and third Brillouin zones in the reduced zone scheme. The sections of the second zone in Fig. 6 are put together into a square by translation through an appropriate reciprocal lattice vector. A different G is needed for each piece of a zone.

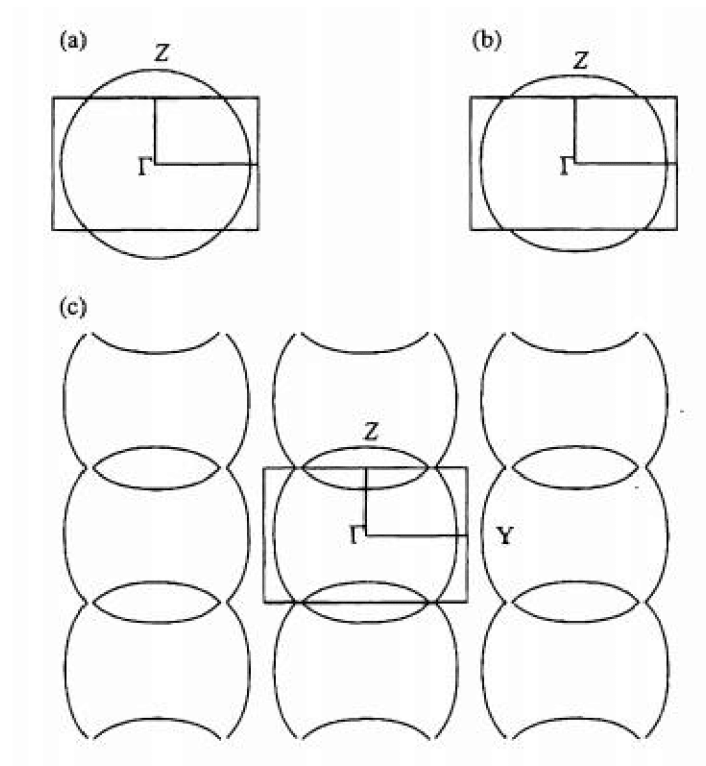
The Fermi Surface is a plane that collects points where the energy of the reciprocal lattice becomes the Fermi level (chemical potential of electrons). If electrons filled in the reverse lattice are present across both the first and second Brillouin zones, a Fermi surface is present and becomes a conductor. If the band gap allows the electrons to fill the first Brillouin zone and fail to fill the second Brillouin zone, the Fermi surface does not exist and becomes a nonconductor.

At absolute zero, its momentum space is filled with the lowest momentum. According to the Pauli Beta principle, electrons cannot have the same state. Therefore, you have to fill in the lowest exercise amount one by one. Let's say there are n electrons without considering the spin, and we must occupy n states in the k -space. However, since the arrangement at the lowest temperature must be satisfied, all states must be stacked sequentially from the lowest state to the top. Eventually, it is made like a surface, and there is a distinction between the state with electrons and the state without electrons, but beyond that surface, there is no electron.

Therefore, electrons exist only on the surface and inside the surface. The Brillouin zone is a simple lattice, but the Fermi surface is always present with or without the Brillouin zone. if there are few electron states in the Fermi surface, it is expressed only in the first Brillouin zone.

The Fermi surfaces of many simple di- and trivalent metals can be understood adequately by the following sequence of processes.

- (1) Construct a free-electron Fermi sphere corresponding to the number of valence electrons.
- (2) Construct a sufficient number of Brillouin zones to enclose the Fermi sphere.
- (3) Split and 'round off' the edges of the Fermi surface wherever it cuts a Brillouin zone boundary (i.e. points at which band gaps opens up)
- (4) Apply the periodicity of k -space by replicating all of the Fermi surface sections at equivalent points in the first Brillouin zone.



The evolution of the Fermi surface of a divalent two-dimensional metal as a bandgap is opened at the Brillouin zone boundary.

- (a) is free-electron Fermi circle and rectangular Brillouin zone;
- (b) is the effect of a band-gap opening up at the Brillouin zone boundary
- (c) is resulting Fermi surface sections in the extended zone scheme.

2. The nearly-free-electron model

2.1 Perturbation Theory

In mathematics and applied mathematics, perturbation theory comprises methods for finding an approximate solution to a problem, by starting from the exact solution of a related, simpler problem. A critical feature of the technique is a middle step that breaks the problem into "solvable" and "perturbative" parts. In perturbation theory, the solution is expressed as a power series in a small parameter ε . The first term is the known solution to the solvable problem. Successive terms in the series at higher powers of ε usually become smaller. An approximate 'perturbation solution' is obtained by truncating the series, usually by keeping only the first two terms, the solution to the known problem and the 'first order' perturbation correction.

Perturbation theory is used in a wide range of fields, and reaches its most sophisticated and advanced forms in quantum field theory. Perturbation theory (quantum mechanics) describes the use of this method in quantum mechanics. The field in general remains actively and heavily researched across multiple disciplines.

In quantum mechanics, perturbation theory is a set of approximation schemes directly related to mathematical perturbation for describing a complicated quantum system in terms of a simpler one. The idea is to start with a simple system for which a mathematical solution is known, and add an additional "perturbing" Hamiltonian representing a weak disturbance to the system. If the disturbance is not too large, the various physical quantities associated with the perturbed system (e.g. its energy levels and eigenstates) can be expressed as "corrections" to those of the simple system. These corrections, being small compared to the size of the quantities themselves, can be calculated using approximate methods such as asymptotic series. The complicated system can therefore be studied based on knowledge of the simpler one. In effect, it is describing a complicated unsolved system using a simple, solvable system.

Time-independent perturbation theory

Time-independent perturbation theory is one of two categories of perturbation theory, the other being time-dependent perturbation (see next section). In time-independent perturbation theory, the perturbation Hamiltonian is static (i.e., possesses no time dependence).

What we need in this chapter is ‘First order corrections’ and ‘Second order corrections’, which are very complicated, so if we omit them and describe only the results as follows.

$$|n^{(1)}\rangle = \sum_{k \neq n} \frac{\langle k^{(0)} | V | n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} |k^{(0)}\rangle.$$

and up to second order, the expressions for the energies and eigenstates are :

$$\begin{aligned} E_n(\lambda) &= E_n^{(0)} + \lambda \langle n^{(0)} | V | n^{(0)} \rangle + \lambda^2 \sum_{k \neq n} \frac{|\langle k^{(0)} | V | n^{(0)} \rangle|^2}{E_n^{(0)} - E_k^{(0)}} + O(\lambda^3) \\ |n(\lambda)\rangle &= |n^{(0)}\rangle + \lambda \sum_{k \neq n} |k^{(0)}\rangle \frac{\langle k^{(0)} | V | n^{(0)} \rangle}{E_n^{(0)} - E_k^{(0)}} + \lambda^2 \sum_{k \neq n} \sum_{\ell \neq n} |k^{(0)}\rangle \frac{\langle k^{(0)} | V | \ell^{(0)} \rangle \langle \ell^{(0)} | V | n^{(0)} \rangle}{(E_n^{(0)} - E_k^{(0)}) (E_n^{(0)} - E_\ell^{(0)})} \\ &\quad - \lambda^2 \sum_{k \neq n} |k^{(0)}\rangle \frac{\langle k^{(0)} | V | n^{(0)} \rangle \langle n^{(0)} | V | n^{(0)} \rangle}{(E_n^{(0)} - E_k^{(0)})^2} - \frac{1}{2} \lambda^2 |n^{(0)}\rangle \sum_{k \neq n} \frac{|\langle k^{(0)} | V | n^{(0)} \rangle|^2}{(E_n^{(0)} - E_k^{(0)})^2} + O(\lambda^3). \end{aligned}$$

2.2 The Schrödinger equation in a periodic potential with Hamiltonian

Let's think about why we learn about the near-free electron model. While studying this part, I thought about why we should learn about this near-free electron. Previously, we had XPS experiments, when we could see graphs drawn on what crystal structures mean, but we remember that we could not visualize the meaning and crystal structures in parentheses.

First, we learned Bloch's theorem in 2 chapter. To sum it up briefly, Bloch's Theorem : An electron in a periodic potential has eigenstates of the form,

$$\psi_k^\alpha(r) = e^{ik \cdot r} u_k^\alpha(r)$$

where u_k^α is periodic in the unit cell and k (the crystal momentum) can be chosen within the first Brillouin zone. index alpha is band index.

We start with completely free electrons whose Hamiltonian is

$$H_0 = \frac{p^2}{2m} \text{ (Completely free electrons)}$$

The corresponding energy eigenstates, the plane waves $|k\rangle$, have eigenenergies given by

$$\varepsilon_0(k) = \frac{\hbar^2 |k|^2}{2m} \text{ (Eigenenergy)}$$

Now consider a weak periodic potential perturbation to this Hamiltonian

$$H = H_0 + V(r) \text{ (A periodic potential perturbation 1)}$$

with V periodic, meaning

$$V(r) = V(r + R) \text{ (A periodic potential perturbation 2)}$$

where R is any lattice vector. The matrix elements of this potential are then just the Fourier components

$$\langle k' | V | k \rangle = \frac{1}{L^3} \int dr e^{i(k-k') \cdot r} V(r) \equiv V_{k'-k} \dots\dots\dots [1]$$

which is zero unless $k' - k$ is a reciprocal lattice vector. Thus, any plane-wave state k can scatter into another plane-wave state k' only if these two plane waves are separated by a reciprocal lattice vector. if two plane waves are separated by a reciprocal latticed vector, any plane-wave stake k and sactter into another plane-wave state k' .

We now apply the rules of perturbation theory. At first order in the perturbation V , we have

$$\varepsilon(k) = \varepsilon_0(k) + \langle k | V | k \rangle = \varepsilon_0(k) + V_0$$

which is just an uninteresting constant energy shift to all of the eigenstates.

In fact, it is an exact statement (at any order of perturbation theory) that the only effect of V_0 is to shift the energies of all of the eigenstates by this constant.¹ Henceforth we will assume that V_0 is constant and $V_0 = 0$ for simplicity.

At second order in perturbation theory we have

$$\varepsilon(k) = \varepsilon_0(k) + \sum_{k' = k + G} ', \frac{|\langle k' | V | k \rangle|^2}{\varepsilon_0(k) - \varepsilon_0(k')} \dots\dots\dots [2]$$

Since the Fourier mode at $G = 0$ is just a constant independent of position.

where the ' means that the sum is restricted to have $G \neq 0$. In this sum, however, we have to be careful. It is possible that, for some k' it happens that $\varepsilon_0(k)$ is very close to $\varepsilon_0(k')$ or perhaps they are even equal, in which case the corresponding term of the sum diverges and the perturbation expansion makes no sense. This is what we call a degenerate situation and it needs to be handled with degenerate perturbation theory, which we shall consider in a moment.

To see when this degenerate situation occurs, we look for solutions of the two equations

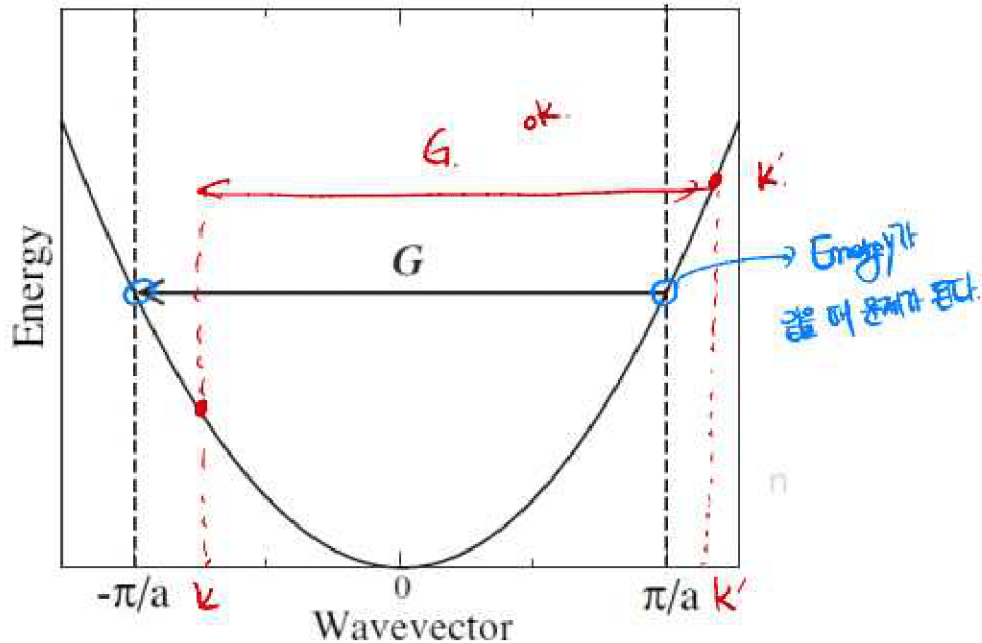
$$\varepsilon_0(k) = \varepsilon_0(k') \dots\dots\dots [3]$$

$$k' = k + G \dots\dots\dots [4]$$

First, let us consider the one-dimensional case. Since $\varepsilon(k) \sim k^2$, the only possible solution of [3] is $k' = -k$. This means the two equations are only satisfied for

$$k' = -k = \frac{n\pi}{a}$$

or precisely on the Brillouin zone boundaries (see Fig. below).



This happens also in higher dimensions

Scattering from Brillouin zone boundary to Brillouin zone boundary. The states at the two zone boundaries are separated by a reciprocal lattice vector G and have the same energy.

This situation leads to a divergence in perturbation theory, [2], because when the two energies match, the denominator is zero.

In fact, this is quite general even in higher dimensions: given a point k on a Brillouin zone boundary, there is another point k' (also on a Brillouin zone boundary) such that [3] and [4] are satisfied.

When we are very near a zone boundary, since [2] is divergent, we need to handle this situation with degenerate perturbation theory. In this approach, one diagonalizes the Hamiltonian within the degenerate space first (and other perturbations can be treated after this). In other words, we take states of the same energy that are connected by the matrix element and treat their mixing exactly.

2.3 Degenerate State Perturbation Theory

The perturbation theory is a tool to mathematically solve how the exact solution changes when it makes a minute change to a well-known system throughout physics.

The perturbation expansion has a problem for states very close in energy. The energy difference in the denominators goes to zero and the corrections are no longer small. The series does not converge. We can very effectively solve this problem by treating all the (nearly) degenerate states like we did ϕ_n in the regular perturbation expansion. That is, the zeroth order state will be allowed to be an arbitrary linear combination of the degenerate states and the eigenvalue problem will be solved.

Assume that two or more states are (nearly) degenerate. Define N to be the set of those nearly degenerate states. Choose a set of basis state in N which are orthonormal $\langle \phi^{(j)} | \phi^{(i)} \rangle = \delta_{ji}$ where i and j are in the set N . We will use the indices i and j to label the states in N .

By looking at the zeroth and first order terms in the Schrödinger equation and dotting it into one of the degenerate states $\phi^{(j)}$, we derive the energy equation for first order (nearly) degenerate state perturbation theory

$$\sum_{i \in N} \langle \phi^{(j)} | H_0 + H_1 | \phi^{(i)} \rangle \alpha_i = E \alpha_j,$$

This is an eigenvalue equation with as many solutions as there are degenerate states in our set.

We recognize this as simply the (matrix) energy eigenvalue equation limited the list of degenerate states. We solve the equation to get the energy eigenvalues and energy eigenstates, correct to first order.

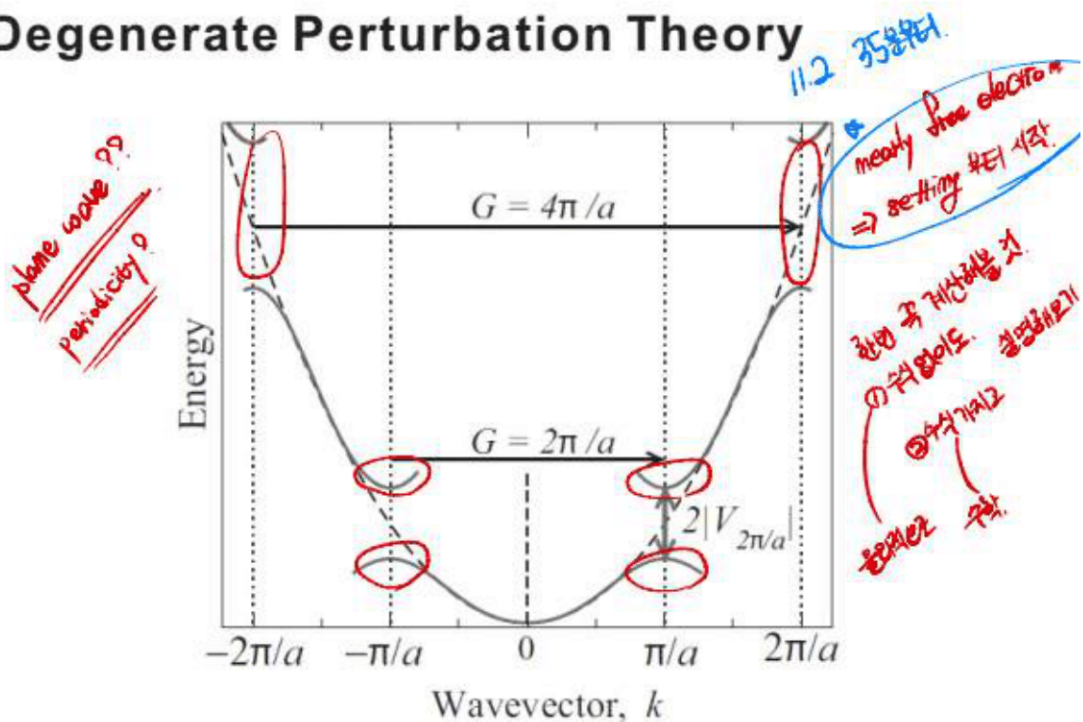
Written as a matrix, the equation is

$$\begin{pmatrix} H_{11} & H_{12} & \dots & H_{1n} \\ H_{21} & H_{22} & \dots & H_{2n} \\ \dots & \dots & \dots & \dots \\ H_{n1} & H_{n2} & \dots & H_{nn} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_n \end{pmatrix} = E \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \dots \\ \alpha_n \end{pmatrix}$$

where $H_{ji} = \langle \phi^{(j)} | H_0 + H_1 | \phi^{(i)} \rangle$ is the matrix element of the full Hamiltonian. If there are n nearly degenerate states, there are n solutions to this equation.

The Stark effect for the (principle quantum number) $n=2$ states of hydrogen requires the use of degenerate state perturbation theory since there are four states with (nearly) the same energies. For our first calculation, we will ignore the hydrogen fine structure and assume that the four states are exactly degenerate, each with unperturbed energy of E_0 . That is $H_0 \phi_{2lm} = E_0 \phi_{2lm}$. The degenerate states ϕ_{200} , ϕ_{211} , ϕ_{210} , and $\phi_{21(-1)}$.

Degenerate Perturbation Theory



To explain the above concept more directly, if two plane-wave states $|k\rangle$ and $|k'\rangle = |k+G\rangle$ are of approximately the same energy (meaning that k and k' are close to zone boundaries), then we must diagonalize the matrix elements of these states first. We have

$$\begin{aligned}
\langle k | H | k \rangle &= \epsilon_0(k) \\
\langle k' | H | k' \rangle &= \epsilon_0(k') = \epsilon_0(k + G) \\
\langle k | H | k' \rangle &= V_{k-k'} = V_G^* \\
\langle k' | H | k \rangle &= V_{k'-k} = V_G
\end{aligned}$$

$V_{-G} = V_G^*$ is guaranteed by the fact that $V(r)$ is real.

$$|\psi\rangle = \alpha|k\rangle + \beta|k'\rangle = \alpha|k\rangle + \beta|k+G\rangle$$

In one-dimension,

$$V(x) = \tilde{V} \cos(2\pi x/a)$$

$$|\Psi_{\pm}\rangle = \frac{1}{\sqrt{2}}(|k\rangle \pm |k'\rangle)$$

3. Vanishing potential

The following equation was derived using $V_0 = 0$ as a simple equation calculation of introduction.

$$(E_{q-G}^0 - E)C_{q-G} + \sum_{G''} V_{G''-G} C_{q-G''} = 0$$

Here, i will name this equation as 'It' in this concept.

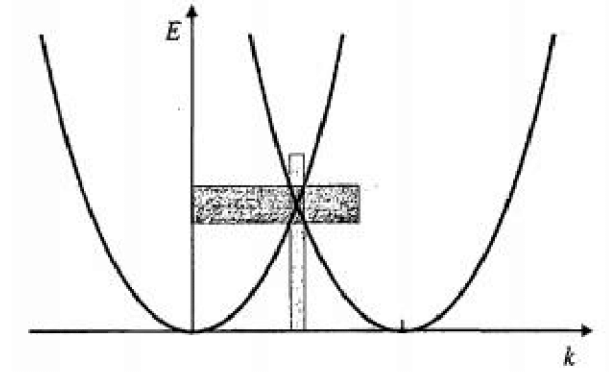
3.1 Single electron energy state

This just means that there are no other states at the same energy at that point in k-space. Thus, we take 'it' and let $V_G \rightarrow 0$, giving

$$(E_{q-G}^0 - E)C_{q-G} = 0 \cdots (i)$$

$$E_{q-G} - E = 0 \cdots (ii)$$

$$C_{q-G} = 0 \cdots (iii)$$



But we need to define that $C_{q-G'}$ must be finite for this state and zero for all other cases, because of (ii)

$$E(q - G') = E_{q-G'}^0 = \frac{\hbar^2 (q - G')^2}{2m}$$

$$\psi \propto e^{i(q-G') \cdot r}$$

In other words, the solution is a single plane wave. However, there is another, much more important implication; note that $C_{q-G} = 0$ for $G \neq G'$ as $V_G \rightarrow 0$, implies that the sum term of 'it's' ex-form only causes corrections to the energy which are second order in the potential V when it is finite but small.

Therefore, in the case of well-separated free-electron states and a weak potential, 'it' can be rewritten

$$(E_{q-G'}^0 - E)C_{q-G'} + O(V^2) = 0$$

3.2 Several degenerate energy levels

Let us now assume that several different free-electron dispersion relationships cross at a particular point in k -space. (ii) can be true simultaneously for several different levels. This means that a number of $C_{q-G'}$ can be arbitrarily large and the wavefunction will become a linear combination of plane waves.

3.3 Two degenerate free-electron levels

Take two dispersion curves $E_{q-G_1}^0$ and $E_{q-G_2}^0$ at a point in k -space where their energy separation is less than $|V|$,

$$|E_{q-G_1}^0 - E_{q-G_2}^0| \leq |V|$$

For all other $G \neq G_2$, however,

$$(E - E_{q-G_1}^0)C_{q-G_1} = V_{G_1-G_1}C_{q-G_1} + V_{G_2-G_1}C_{q-G_2} + \sum_{G \neq G_1, G_2} V_{G-G_1}C_{q-G} \dots\dots\dots(1)$$

and

$$(E - E_{q-G_2}^0)C_{q-G_2} = V_{G_2-G_2}C_{q-G_2} + V_{G_1-G_2}C_{q-G_1} + \sum_{G \neq G_1, G_2} V_{G-G_2}C_{q-G} \dots\dots\dots(2)$$

As before, $V_0 \equiv V_{G_1-G_1} \equiv V_{G_2-G_2} \equiv 0$ and ignoring the small $\sum$ terms. besides, $k = q - G_1$, $G = G_2 - G_1$ are substituted.

hence, the (1) equation and (2) equation are simultaneously simplified as,

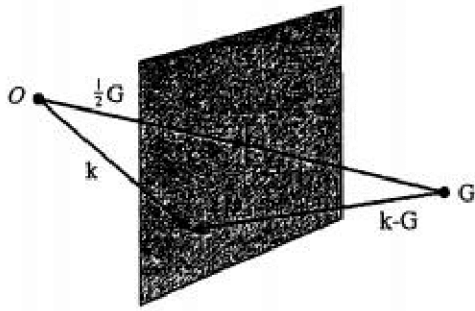
$$(E - E_k^0)C_k = V_G C_{k-G} \dots\dots\dots (3)$$

and

$$(E - E_{k-G}^0)C_{k-G} = V_{-G} C_k = V_G^* C_k \dots\dots\dots (4)$$

where we have used the fact that the potential must be real to give $V_{-G} = V_G^*$. (3) and (4) are simultaneous equations of coefficients C_k and C_{k-G} ; the solution is

$$E \approx \frac{1}{2}(E_k^0 + E_{k-G}^0) \pm |V_G|.$$

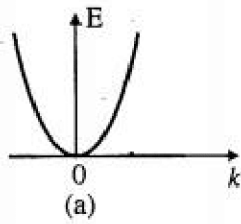


Therefore we have a symmetrical splitting of levels about the free electron value close to where E_k^0 and E_{k-G}^0 cross. Now $E_k^0 = E_{k-G}^0$ when $|k| = |k - G|$. In other words, k lies on a plane half way between 0 and G . The vector $k - \frac{1}{2}G$ will be parallel to this plane, so that $G \cdot (k - \frac{1}{2}G) = 0$, i.e.

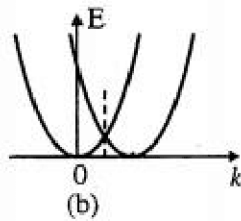
$$G^2 = 2k \cdot G \dots\dots\dots (\clubsuit)$$

We can say that a weak periodic potential has its stronger effect on states with wavevectors close to the conditions defined by $(\clubsuit)$.

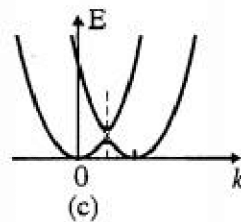
These are the evolution of nearly-free-electron bands:



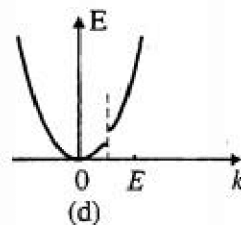
(a) is initial free-electron dispersion relationship;



(b) is replication of free-electron dispersion curves due to periodicity of k -space;

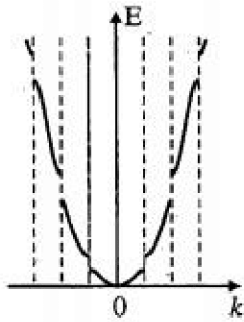


(c) is opening up at a band gap at the point on the Brillouin-zone boundary where two free-electron curves cross;

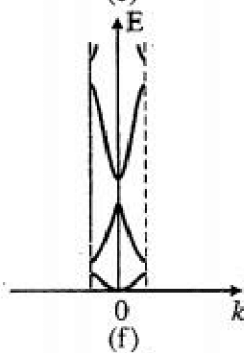


(d) is the effect on the free-electron dispersion relationship of a band gap opening at each successive Brillouin zone boundary. The periodicity of k -space means that the curves shown in (d) must be present in every Brillouin zone;

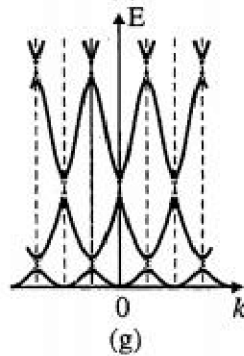
These show different ways of representing the resulting bandstructure:



(e) is the extended-zone scheme, in which each Brillouin zone has one band shown;



(f) is the reduced-zone scheme whereby all bands are shown within the first Brillouin zone;



(g) is the repeated-zone or periodic-zone scheme.